

UNIVERSITY OF BIRMINGHAM

Fairness and Bias in Intelligent Interactive Systems

A Review of Algorithmic Retention and High-Stimulant Bias in Short-Form Social Platforms

Ilya Sullivan

Submitted: February 2026

Index

1 Introduction
    1.1 Research Questions
2 Method
    2.1 Search Strategy
3 Summary of Research
    3.1 The Role of Engagement Metrics in Content Bias
    3.2 Mechanisms of Retention and Algorithmic Drift
        3.2.1 Feedback Loops
        3.2.2 Echo Chambers
        3.2.3 Dark Patterns
    3.3 Review of Mitigation Strategies
4 Conclusions
5 References

Abstract

Short-form video (SFV) platforms have become one of the leading forms of entertainment and information consumption, yet the recommendation algorithms behind them still remain unoptimised to reduce bias and ensure videos shown to users have a fair distribution. This systematic literature review examines the different ways in which bias is created including from “engagement first” goal strategies, reduced metrics and content engagement disparity. There are three main dimensions of algorithmic bias that are analysed, including feedback loops, echo chambers and dark patterns, all of which constrain user autonomy. Mitigation strategies are analysed to reduce these effects including Constrained Markov Decision Processes in reinforcement learning and antidote data injection.

Keywords: Recommender System, Intelligent Interactive System, Short-Form Video (SFV), Bias, Fairness, Engagement-First Optimisation, Feedback Loop, Echo Chamber, Dark Pattern

1 Introduction

In the current day, Short-Form Video (SFV) content has increasingly become more prevalent in everyone's lives, now with many companies using it for not only advertising but for the latest news and information , requiring many who would not normally watch short-form content, to watch it to stay up to date [1]. Many of these social shifts have been accelerated due to COVID-19 [2], as well as the rapid adoption from younger generations with ages 4-34 having social media taking the majority of daily media consumption time, as per Ofcom's Media Nations UK Review, 2025 [3]; on average, adults spend 51 minutes per day on social media as of 2025, up from 47 minutes in 2024 [3].

In this literature review, we will explore the fairness and bias in the recommendation algorithms of SFV platforms, specifically Instagram Reels, YouTube Shorts and TikTok, and investigate the ways in which they increase watch-time and retain users, by using engagement-first optimisation strategies over algorithms that optimise for user wellbeing. Additionally, we will analyse the effects of how algorithmic retention methods affect people and ways in which they can be fixed to allow users to have more autonomy with their viewing habits.
The context of fairness in the context of this paper refers to the equal distribution of content types presented to the user, whilst bias refers to the systematic prioritisation of content which is "engagement-first", including high-intensity and high-arousal content, aiming to attract the most, likes, high watch time percentage, reposting, sharing and/or commenting.

1.1 Research Questions

For the purposes of ensuring that we had covered the scope of research, three questions were conceptualised:

- **RQ1**: How do "Engagement-First" recommendation algorithms create bias towards high-intensity content over educational or low-intensity content?
- **RQ2**: To what extent do current recommendation algorithms anchor users in recursive feedback loops and what role does "algorithmic drift" play in limiting diversity for users?
- **RQ3**: What design interventions can mitigate the intensity of feedback loops to support intentional user choice?

2 Method

2.1 Search Strategy

A variety of different sources were chosen throughout the process. Initial research was conducted through Google Scholar in order to find related papers to the area of bias in short-form content. For research into specific topics, ACM Digital Library and Springer Nature were selected to get the most accurate information. In order to ensure relevancy to the short-form content post-COVID-19, all papers selected were published from between January 2023 and January 2026. The following query was used to find relevant papers:

(*"short-form video"* OR *"TikTok"* OR *"YouTube Shorts"* OR *"Instagram Reels"*) OR (*"recommender system"* OR *"recommendation algorithm"*) AND (*"bias"* OR *"fairness"* OR *"filter bubble"* OR *"echo chamber"* OR *"feedback loop"* OR *"dark pattern"* OR *"engagement"*)

To screen through the papers, the PRISMA method was used, using the following ruleset: Papers must be peer reviewed, published >01/2022, on "YouTube Shorts" and/or "TikTok" and/or "Instagram Reels" that discussed bias and fairness within their recommendation systems. Papers found to not be in English, or majoritively on long-form content were excluded.

Database	Results	Selected Papers
Springer Nature	18,217	[1, 12, 13, 16]
ACM	8,766	[2, 8, 9, 14, 17, 18, 19, 20]
Google Scholar	7,740	[3, 5, 6, 7, 10, 11, 15]

3 Summary of Research

3.1 The Role of Engagement Metrics in Content Bias

Recommendation algorithms (recommender systems [4]) are integrated into much of society, from tailored notifications in the morning, to picking where to go, to choosing what to make for dinner, the primary aim, to give the most relevant result to satisfy the user [4]. In order to successfully judge relevancy, typical video platforms use interactions such as clicks, purchases, favourites. However, with short-form video, metrics differ, basing recommendations on more subtle cues such as watch-time, click-through rate as well as comment engagement, likes, shares, and others [5], setting up the foundation for each platform's own recommendation algorithm.

The problem which all these algorithms face is a natural result of the biases caused by basing the majority of the recommendations off these metrics; The end product would favour certain types of content that feature high-intensity, high-arousal and/or short videos over others. These types of videos commonly elicit strong emotions such as outrage, fear, anger, which the algorithm interprets as positive feedback regardless of actual engagement type; the result of this is called "engagement-first bias". Content such as educational content is typically low-intensity, where videos are longer and require more focus to pay attention and are focused to a certain audience is favoured less in an interaction-focused algorithm [17].

In traditional long-form video platforms, users engage in querying, having a purpose for being on the platform, whilst SFV platforms on the other-hand are primarily designed for the browsing experience, where vertical scrolling is the key interaction point [2, 5]. This simple action with little-to-no pausing means that a point that was previously measured, choosing one video over another, is no longer easily accessible to the user. This shift in power from user-choice, to algorithm-curated feeds, means that they now must rely on consumption patterns rather than explicit choices to signal preferences [16]. The result is an algorithm in which unchecked accidental or purposeful bias implemented by companies can go undetected.

3.2 Mechanisms of Retention and Algorithmic Drift

3.2.1 Feedback Loops

The process of a "feedback loop" is defined as repeatedly updating recommendations based on interaction data in order to narrow content suggestion [6]. Unlike in Section 3.1, increased retention through feedback loops inherently intensifies recommendations through the accumulation of user data, however are self-limiting rather than infinitely reinforcing [15]; This means that if a user has consistently watched the same type of videos, it is harder to remove similar suggestions.

The dynamics of feedback loops provide critical insight into how algorithmic drift can occur, with the research of YouTube Shorts documenting that topical relevance drops after a single recommendation [12]. This shows there are at least two mechanisms that determine the recommendation: finding similar content; content abandonment factor. In politically sensitive contexts like the South China Sea disputes and Taiwan elections, the shift was immediate, showing that the algorithm prioritizes engagement over thematic coherence [12]. In the context of retention, this implies that even if the user attempts to completely switch topics to something they want to see, the algorithm may switch back to another topic because the user will "engage" with it even if it may not want to see it just because it fits the checkboxes for what counts as engagement.

This evidence is further solidified by a study conducted by Brno University of Technology [8] on YouTube Shorts, finding that recommendations experienced content drift regarding political content. Videos that tested a political outlook would act as seeds to show increasingly more political content in that direction until an equilibrium would be met that the user would engage, maximising the user's attention [8].

It is important to note, these feedback-loops exhibit a front-loaded sensitivity [15], where the user's initial engagement triggers a rapid calibration of the recommendation algorithm. Further consumption continues to refine the current profile but has a marginal difference, until eventually stabilizing. Depending on the type of content, the plateau may slow to different points.

3.2.2 Echo Chambers

A feedback loop acts as the mechanism by which content is refined to get closer to the goal state of finding content closely matching what the user engages with, the effect caused is an echo chamber. These are defined as an environment in which a person encounters only beliefs or opinions that coincide with their own, so that their existing views are reinforced and alternative ideas are not considered.

In a large-scale analysis of TikTok's political landscape, researchers Li, Cheng and Gil de Zúñiga found substantial evidence of filter bubbles and echo chambers on the platform [7]. The group analyzed over 160,000 public accounts on the platform and 16 million videos collected between the dates of 2019 and 2023, and identified significant distinct political clusters of content, increasing on the platform since 2020, likely relating to the US presidential election. One key finding was that right-wing communities were found to be much more isolated from other political networks than left-leaning communities, with far more radical content being tightly connected to each other, acting as an echo-chamber. The study used Granger causality testing to demonstrate that engagement metrics such as likes, comments and shares reinforce subsequent political posting behaviour from the creator [7]. Without this support from the recommender system to promote creator videos, many would typically stop posting or reduce output but this self-reinforcing cycle between algorithmic amplification and user behaviour has the adverse effect of reducing content diversity for the consumer in the long-term.

3.2.3 Dark Patterns

Features designed by SFV platforms further exemplify algorithmic immersion by amplifying the previously mentioned methods. One of the most significant features to this is the infinite scroll interface that discourages natural stopping points such as the slowing of video pace, end of a video and querying a video to watch [9]. With videos being selected for the user, videos lasting under 60 seconds, videos being cached for the chance that the network is unstable, there are minimal reasons as to why a user needs to stop watching other than outside means.

Dark patterns refer to design practices which undermine users' ability to make intentional and informed choices in relation to digital systems [14]. Platforms such as TikTok have applied these sorts of "dark patterns" to keep people on the platform including the user requiring pressing the back button twice to close out of the app — on a single back press, it goes to the next video to hook the user back [10]. Additional mechanisms that are on all the major platforms include having search queries show results but additionally having infinite scroll for those videos as well, slowly moving from the query results to the default tailored feed [10, 11].

3.3 Review of Mitigation Strategies

SFV platforms are so commonly part of the majority of the population's daily life [3] that ensuring their algorithms have reduced bias, increased fairness and diversity of content is becoming increasingly imperative. Feedback loops and filter bubbles currently are the two largest contributors of this type of content bias — these challenges have been recognised as shown in sections 3.1 and 3.2 and solutions have been developed to better shape information consumption.

A common technique used to counteract the effects of bias within recommendation systems is by including antidote data within the training set [17]. Antidote data is developed from an existing pre-trained matrix-factorization-based recommender system and provides the flexibility of

not having to modify the original input data or recommendation algorithm. This would, in-effect, counteract existing biases within the current training data, and unlike data cleaning or reweighting, preserve the original dataset whilst correcting the model [17]. Such examples could include additional users that watch educational SFV to completion in order to counteract the number of people that skip over that type of content reducing the amount that content is recommended to users.

An alternative could be using a Constrained Markov Decision Process (CMDP) in reinforcement learning, dynamically constraining the item exposure fairness metric at each iteration — this has the effect of constraining the reward across multiple metrics, reducing the reward if fairness isn't properly met [17, 18, 19, 20].

4 Conclusions

The conclusions show that within short-form video platforms, the effect of optimising for watch-time by using deep-learning systems, cause heavy bias and reduce the fairness of content distribution, often causing echo chambers of information. The proposed fixes to this include using antidote data or Constrained Markov Decision Processes.

Whilst these SFV platforms have significant latitude in tailoring their algorithms as they see fit, it is clear that many of the practices they have chosen to implement are not fair and are not making progress to change the biases. Instead, they lean into them with dark patterns and additional functions that keep users on the platform, without the consideration of how it affects the user wellbeing and consumption patterns.

5 References

[1] Zhang, P. and Xu, C.K. (2025). How Effective is the Shorts Transformation of Traditional Media? An Analysis from the Perspective of User-Generated Content. Communications in computer and information science, pp.124–137. doi:https://doi.org/10.1007/978-981-96-2373-0_9.

[2] Zannettou, S., Nemes-Nemeth, O., Ayalon, O., Goetzen, A., Gummadi, K., Redmiles, E. and Roesner, F. (2024). Ana-lyzing User Engagement with TikTok's Short Format Video Recommenda-tions using Data Donations. Analyzing User Engagement with TikTok's Short Format Video Recommendations using Data Donations. [online] doi:https://doi.org/10.1145/3613904.3642433.

[3] Ofcom (2025). Media Nations. [online] Available at: https://www.ofcom.org.uk/siteassets/resources/documents/research-and-data/multi-sector/media-nations/2025/media-nations-2025-uk-report.pdf?v=401287.

[4] Ricci, F., Rokach, L. and Shapira, B. (2021). Recommender Systems: Techniques, Applications, and Challenges. Recommender Systems Handbook, pp.1–35. doi:https://doi.org/10.1007/978-1-0716-2197-4_1.

[5] Delft University of Technology (2024). Recommendation Systems of Short Video Platforms: Auditing Algorithms of Short Format Video Platforms to Understand the Rabbit Hole Effect on YouTube Shorts. [online] TUDelft. Available at: https://repository.tudelft.nl/file/File_2759f54b-4982-42c4-b6f0-7fdd8094aa63?preview=1

[6] Lynch, A. (2025). Decoding the Algorithm: The Mathematics Behind TikTok's Short- Form Content SuccessForm Content Success. [online] UCONN Library. Available at: https://digitalcommons.lib.uconn.edu/cgi/viewcontent.cgi?article=2104&context=srhonors_theses.

[7] Li, Y., Cheng, Z. and Gil (2025). TikTok's political landscape: Examining echo chambers and political expression dynamics. New Media & Society. doi:https://doi.org/10.1177/14614448251339755.

[8] Srba, I., Moro, R., Tomlein, M., Pecher, B., Simko, J., Stefancova, E., Kompan, M., Hrckova, A., Podrouzek, J., Gavornik, A. and Bielikova, M. (2022). Auditing YouTube's Recommendation Algorithm for Misinformation Filter Bubbles. ACM Transactions on Recommender Systems, 1(1). doi:https://doi.org/10.1145/3568392.

[9] Violot, C., Elmas, T., Bilogrevic, I. and Humbert, M. (2024). Shorts vs. Regular Videos on YouTube: A Comparative Analysis of User Engagement and Content Creation Trends. arXiv (Cornell University). [online] doi:https://doi.org/10.1145/3614419.3644023.

[10] Mattiuzzo, M. and Paula Pedigoni Ponce (2024). Power through design – dark patterns, personalization and the emergence of TikTok. International Review of Law Computers & Technology, pp.1–25. doi:https://doi.org/10.1080/13600869.2024.2351672.

[11] Jiang, Q. and Ma, L. (2024). Swiping more, thinking less: Using TikTok hinders analytic thinking. Cyberpsychology, 18(3). doi:https://doi.org/10.5817/cp2024-3-1.

[12] Agarwal, N., Cakmak, M.C. and Poudel, D. (2025). Investigating Algorithmic Bias in YouTube Shorts. Lecture Notes in Social Networks, pp.247–261. doi:https://doi.org/10.1007/978-3-032-13509-4_19.

[13] Cakmak, M.C., Agarwal, N. and Oni, R. (2024). The bias beneath: analyzing drift in YouTube's algorithmic recommendations. Social Network Analysis and Mining, 14(1). doi:https://doi.org/10.1007/s13278-024-01343-5.

[14] Ahuja, S., Gunawan, J., Nataliia Bielova and Santos, C.T. (2025). Towards Key Contributing Factors in Identifying Dark Pattern Autonomy Violations under the EU Digital Services Act. pp.501–507. doi:https://doi.org/10.1145/3715668.3736336.

[15] Trilling, D. (2024). Communicative Feedback Loops in the Digital Society. Weizenbaum Journal of the Digital Society, [online] 4(2). doi:https://doi.org/10.34669/wi.wjds/4.2.4.

[16] Carlos, J. and Lara, F. (2023). AI-powered recommender systems and the preservation of personal autonomy. AI & society. doi:https://doi.org/10.1007/s00146-023-01720-2.

[17] Li, Y., Chen, H., Xu, S., Ge, Y., Tan, J., Liu, S. and Zhang, Y. (2023). Fairness in Recommendation: Foundations, Methods and Applications. ACM Transactions on Intelligent Systems and Technology. doi:https://doi.org/10.1145/3610302.

[18] Jin, D., Wang, L., He, Z., Zheng, Y., Ding, W., Xia, F. and Pan, S. (2023). A survey on fairness-aware recommender systems. Information Fusion, 100, pp.101906–101906. doi:https://doi.org/10.1016/j.inffus.2023.101906.

[19] Ge, Y., Liu, S., Gao, R., Xian, Y., Li, Y., Huang, X.-J., Pei, C., Sun, F., Ge, J., Ou, W. and Zhang, Y. (2021). Towards Long-term Fairness in Recommendation. arXiv (Cornell University). doi:https://doi.org/10.1145/3437963.3441824.

[20] Ge, Y., Zhao, X., Yu, L., Paul, S., Hu, D., Hsieh, C.-C. and Zhang, Y. (2022). Toward Pareto Efficient Fairness-Utility Trade-off in Recommendation through Reinforcement Learning. Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining. doi:https://doi.org/10.1145/3488560.3498487.